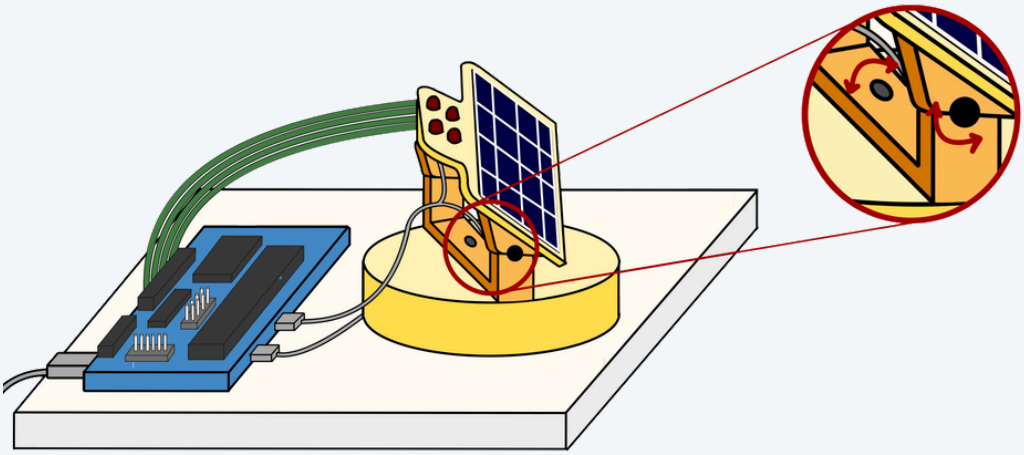
FHILIA WIJAYA

Automated Solar Tracking System



SunSync

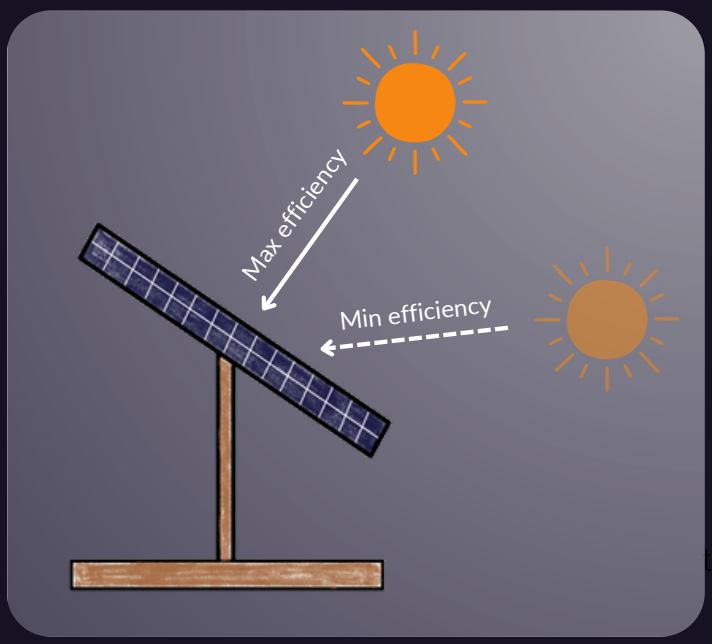
Project Context

This project was developed for ENGR1000J: Introduction to Engineering, where students were tasked with designing a prototype that demonstrates a practical engineering solution.

Our team explored how solar energy systems could improve efficiency by automatically adjusting panel orientation throughout the day.

The goal was to build a small-scale solar tracking system capable of detecting light direction and adjusting the panel position to maximuze energy exposure

Problem



Solar panels installed in a fixed position cannot continuously face the sun as it moves across the sky. This causes energy loss because the panel is not always aligned with the strongest light source. In fact, fixed=mount systems can lose up to 30% of potential energy output due to inefficient orientation. This project explores whether a simple automated tracking system could improve solar panel efficiency.

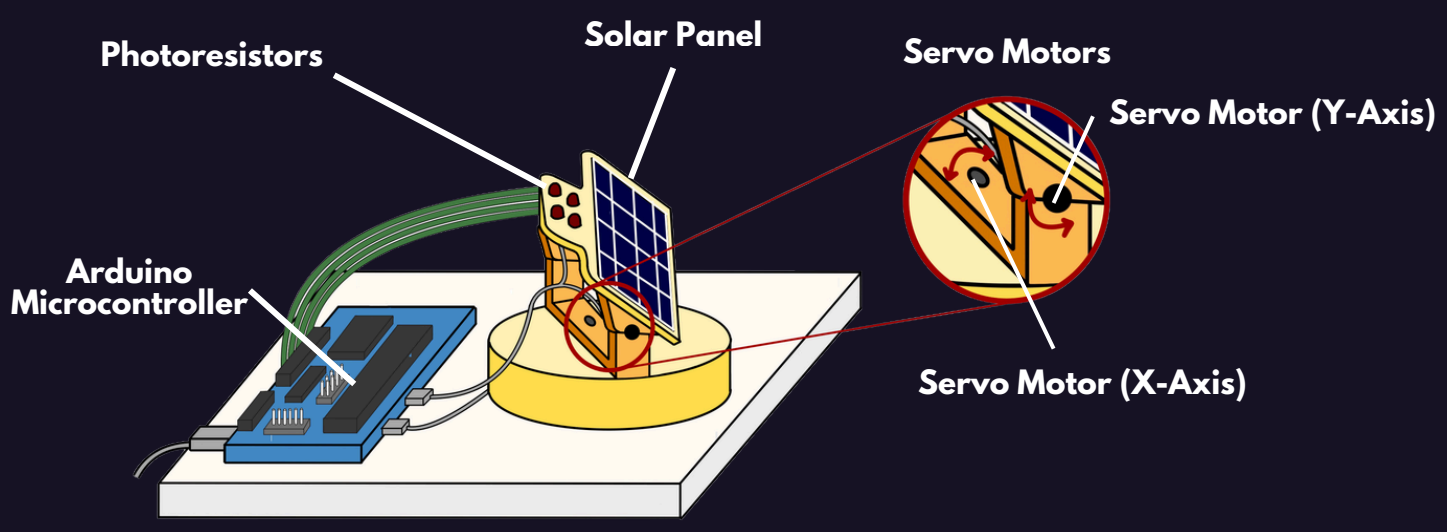
Solution Concept

We designed a sun-tracking solar panel prototype that automatically adjusts its orientation based on light intensity.

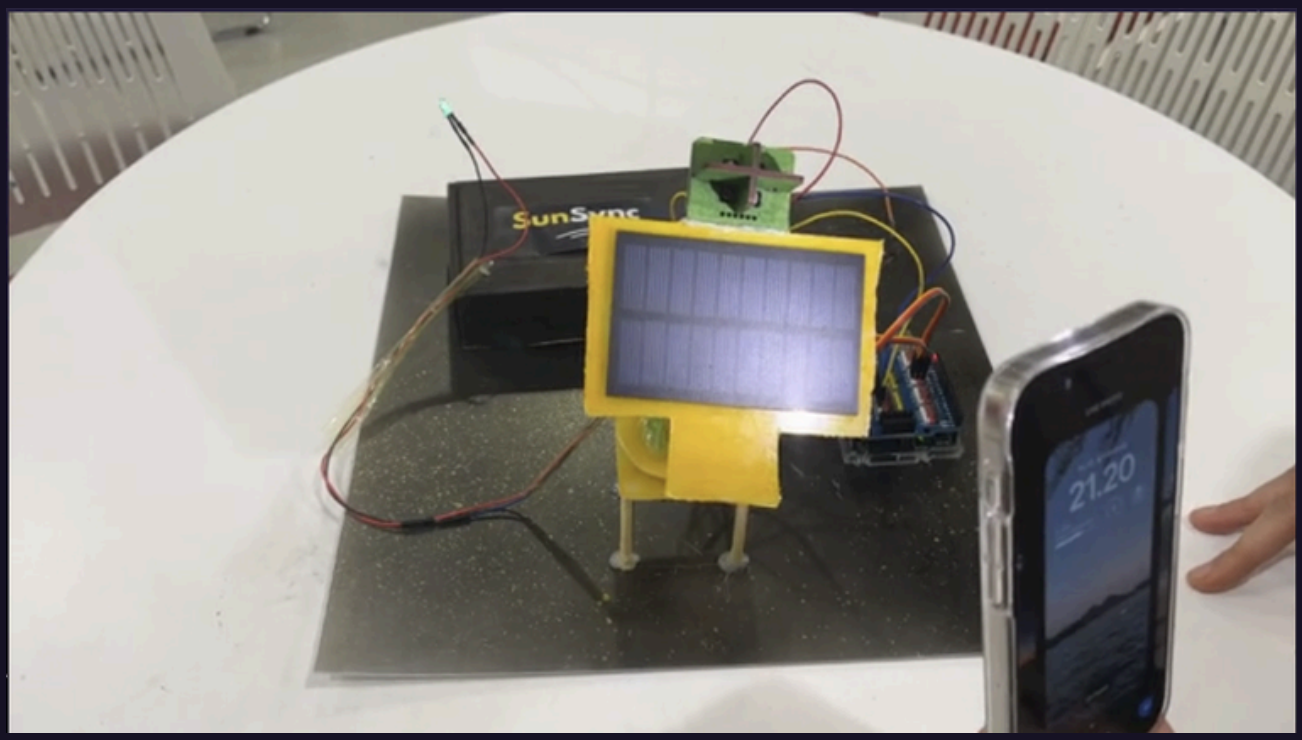
The system uses:

- light sensors (photoresistors)
- Arduino microcontroller
- Servo motors

By comparing light levels on different sensors, the system determines where the strongest light sources is and rotates the panel accordingly.



Prototype

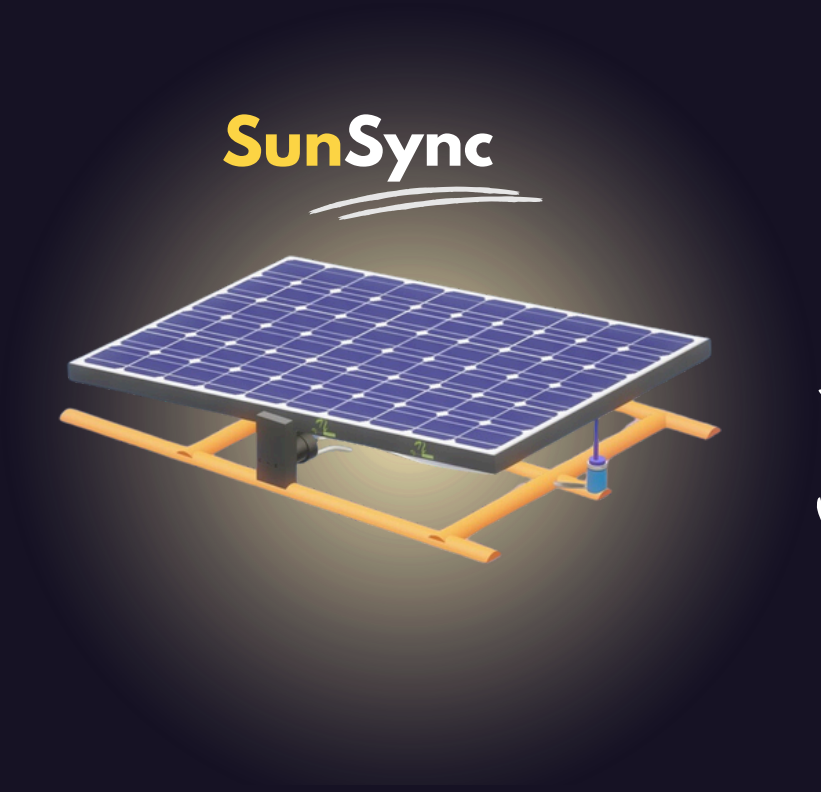


Result

Testing compaared the efficiency of a fixed-mount panel with the tracking system. Results showed the tracking system maintained higher efficiency throughout the day, reaching around 25% compared to 16% of fixed systems. This demonstrates the potential benefit of dynamic solar panel alignment.

Reflection

Working on this project helped me to understand how hardware and software come together in a real system. I learned how sensors, microcontrollers, and motors can interact to automatically respond to environmental changes. It also showed me that engineering design isn't only about making something work, but also about considering efficiency, cost, and practicality.



01 Improved Efficiency

02 Increased Energy Output

03 Better Performances